

Article Title

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

Keywords: Adult neurogenesis, hippocampus, computational model, plasticity, dentate gyrus, CA3

MSC Classification: 92C20

1 Introduction

The Introduction section, of referenced text expands on the background of the work (some overlap with the Abstract is acceptable). The introduction should not include subheadings.

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version), including coloured font. These will not be replicated in the typeset article if it is accepted.

2 Methods

2.1 Network architecture

The DG-CA3 network architecture, shown in Fig. 1, was built primarily upon the models of Kopsick et al. (2024); Kim and Lim (2024a); Yang et al. (2025); Chavlis et al. (2017) and reproduced at a scale of $\frac{1}{500}$ of the rat hippocampus, the per-population cell counts being listed in Table 1. Each neuron was reduced to a single compartment, with no explicit dendrites or axons, yet kept biologically plausible: its neuron and synapse models, together with the cell counts and connectivity, were adapted from Hippocampome.org, an open-access knowledge base that consolidates multiple experimental sources on the hippocampal formation (Wheeler et al. 2023).

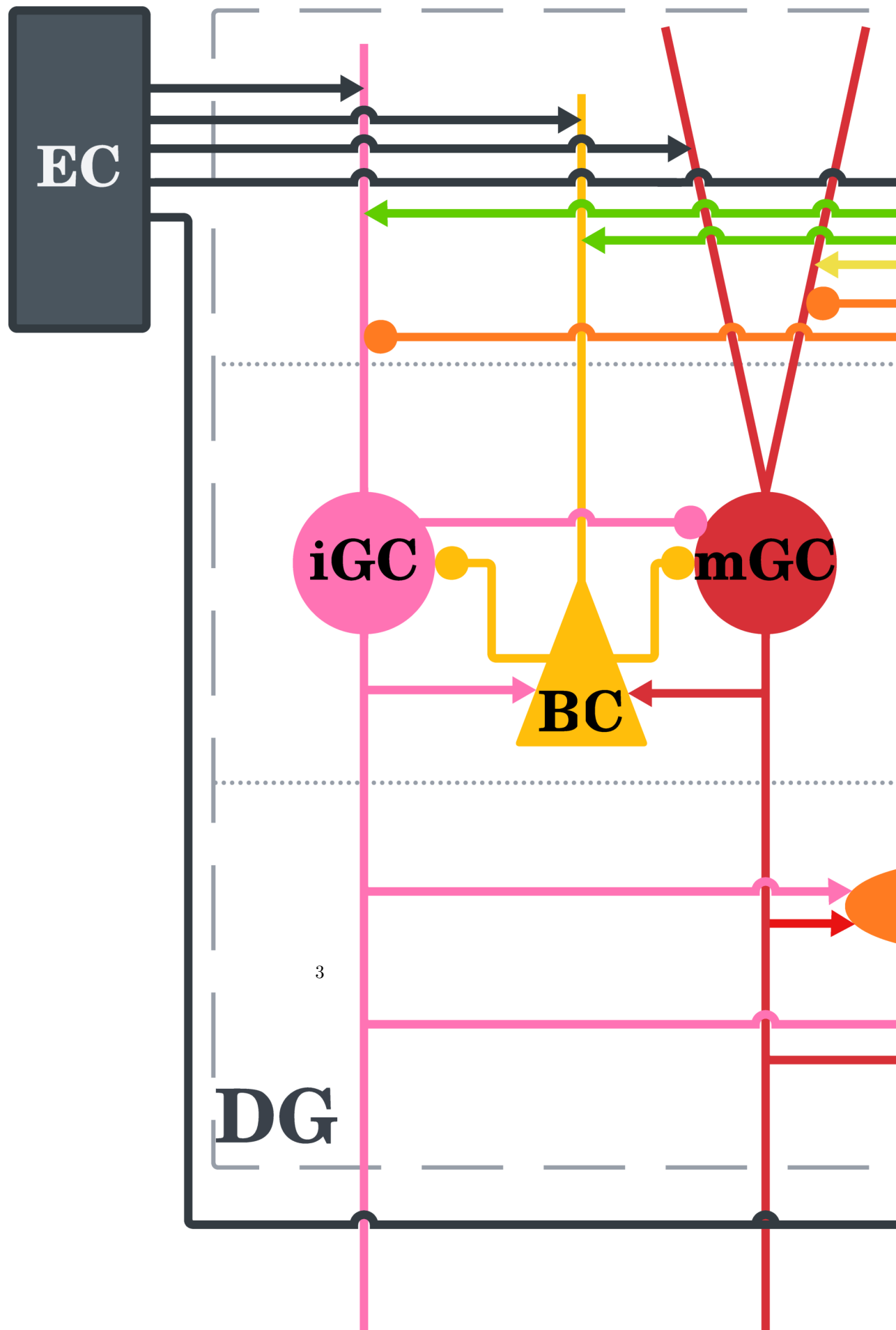
The model included the core excitatory and inhibitory cell types of the DG-CA3 network underlying pattern separation and completion. Although both regions contain many additional interneuron subtypes, several of which are also characterized in Hippocampome.org (Wheeler et al. 2023), this reduced set has been shown in previous computational models to be sufficient to support pattern separation (Chavlis et al. 2017; Kim and Lim 2024b; Yang et al. 2025). The synapses between cell types and their connection probabilities were likewise drawn from multiple sources and kept within biologically reasonable ranges. Once all parameters had been gathered from Hippocampome.org and the remaining literature and the network had been assembled, a subset of neuron and synapse parameters, as well as some connection probabilities, was tweaked within reasonable bounds so that each cell type matched its *in vivo* firing rate and population sparsity (Online Resource 1).

The network input is composed of cells from the entorhinal cortex, with a total of $N_{EC} = 400$ neurons (Amaral et al. 1990; Kim and Lim 2024a). In each simulation, the EC as a whole presents a specific pattern, where each pattern is represented by an active subpopulation of 10% of EC neurons (McNaughton et al. 1991). Inactive neurons not belonging to the pattern do not spike during the simulation, whereas active neurons fire according to a Poisson process with at 40 Hz. EC neurons project their axons via the perforant path to neurons with dendrites in the molecular layer of the DG: GCs, MCs (Scharfman and Myers 2013) and BCs, as well as to CA3 neurons.

The GC population is split into mGCs and iGCs, the latter standing for the cells that adult neurogenesis produces during their characteristic 4–6 week electrophysiological window (Aimone et al. 2014). Of the $N_{GC} = 2000$ GCs (West et al. 1991), 5% are immature (Cameron and McKay 2001), so that $N_{mGC} = 1900$ and $N_{iGC} = 100$. Respecting the lamellar organization of the DG (Sloviter and Lømo 2012), the GCs are distributed across 20 lamellae of 100 cells each. Every GC contacts the BCs and CA3 neurons of its own lamella and, in addition, forms random non-lamellar excitatory connections onto HIPPs and MCs.

iGCs possess distinct electrophysiological characteristics from the mGCs that render them more excitable than mGCs. Furthermore, to simulate their still-incomplete circuit integration, iGCs receive EC afferent inputs at only a fraction $x_{EC \rightarrow iGC}$ of the

Fig. 1 DG-CA3 network architecture. Inhibitory synapses are represented by circles and excitatory ones by arrows. The network input is composed of cells from the EC. The DG has the perforant path and the dendrites of its cells in the molecular layer; the granular layer is composed of iGCs, mGCs and BCs; the hilus is composed of MCs and HIPPs. The CA3 is composed of pCA3 and iCA3 cells.



connection probability between the EC and mGCs, and also receive reduced inhibitory inputs from BCs and HIPPs (Table 3). Additionally, some iGCs are known to also directly inhibit mGCs (Luna et al. 2019), so the model includes an inhibitory synapse from iGCs onto mGCs. Hippocampome.org (Wheeler et al. 2023) does not characterize iGCs. Therefore, the parameters used for iGCs were fitted from the spiking data of Espósito et al. (2005) with the simplex method of Nelder and Mead (1965) (Table 2).

BCs ($N_{BC} = 60$) are inhibitory cells that ensure sparse GC activation through winner-take-all competition among them (Coultrip et al. 1992; Chavlis et al. 2017; Kim and Lim 2024a), where only the most active GCs in a lamella remain active, inhibiting the rest. HIPPs ($N_{HIP} = 60$) also contribute to the sparsity of GC activation, with a more global inhibition acting on all GCs in a lamella. MCs ($N_{MC} = 100$) are excitatory hilar cells that receive excitation from GCs and project back to GCs, BCs, HIPPs and other MCs via cross-lamellar connections. Although these excitatory MC to GC projections exist, their net effect is generally inhibitory through the control of BCs and HIPPs (Myers and Scharfman 2009; Scharfman and Myers 2013).

The HIPP parameters given by Hippocampome.org (Wheeler et al. 2023) reproduce a “rebound burst spiking” behavior, in which a burst of spikes follows the end of a hyperpolarization. Because large hyperpolarizations then drive unbounded bursts that the network cannot tolerate, the HIPP parameters were instead taken from Modak and Chakravarthy (2018).

CA3, unlike the DG, does not follow a lamellar structure in the model, as it forms a far more integrative neural network (Pak et al. 2022; Watson et al. 2025). It holds $N_{pCA3} = 600$ pCA3 cells and $N_{iCA3} = 60$ iCA3 cells, the latter modeled after CA3 basket cells (Wheeler et al. 2023) so as to simplify the immense variety of inhibitory neurons present in CA3 (Kopsick et al. 2024). Both neuronal populations receive afferent inputs from the EC and from GCs, with iCA3 cells inhibiting pCA3 cells. The iCA3 cells inhibit the pCA3 cells, which in turn excite the iCA3 cells and, crucially, wire recurrently among themselves, the substrate of the autoassociation and pattern-completion functions of CA3 (Kopsick et al. 2024; Rolls 2013). A portion of the pCA3 cells further feed back to the DG by contacting MCs, a projection that contributes to pattern separation (Myers and Scharfman 2011).

All simulations were run in Brian2 (Stimberg et al. 2019) with the fourth-order Runge-Kutta method at a fixed time step of 0.1 ms (Butcher 1996); the neuron and synapse parameters appear in Tables 2 and 3. For each input pattern the network was left 300 ms to reach a steady state, and its activity was then recorded over the following 1000 ms. Synaptic connections remained fixed throughout.

2.2 Neuron model

Each neuron obeys the nine-parameter Izhikevich model (Izhikevich 2006, chap. 8). This model was chosen in this work and Hippocampome.org for being capable of reproducing the firing dynamics of neurons across a wide variety of conditions with a biological plausibility approaching that of the Hodgkin-Huxley model (Hodgkin and Huxley 1952), but at a far lower computational cost. Its dynamics are described by:

Cell	N
 EC	400
 mGC	1900
 iGC	100
 MC	100
 HIPP	60
 BC	60
 pCA3	600
 iCA3	60

Table 1 Neuronal population size (N).

$$C_m \frac{dV_m}{dt} = k(V_m - V_r)(V_m - V_t) - u + I \quad (1)$$

$$\frac{du}{dt} = a[b(V_m - V_r) - u] \quad (2)$$

where V_m is the membrane potential, u the recovery variable, C_m the membrane capacitance, V_r the resting potential, V_t the threshold potential, I the total current into the neuron, and k , a and b constants that shape the neuron’s dynamics. These continuous equations are complemented by a discrete rule for action-potential generation (Eq. 3):

$$\text{if } V_m \geq V_{\text{peak}}, \quad \begin{cases} V_m \leftarrow V_{\text{min}} \\ u \leftarrow u + d \end{cases} \quad (3)$$

When V_m reaches the peak value V_{peak} , a spike is emitted, V_m is reset to the after-spike potential V_{min} , and the recovery variable is incremented by d , which makes the following spikes harder to elicit.

2.3 Synapse model

The synapse model follows the formulation of [Senn et al. \(2001\)](#) and [Mongillo et al. \(2008\)](#). It captures short-term depression, from neurotransmitter depletion, and facilitation, from calcium accumulation, acting over hundreds of milliseconds. Five parameters describe each synapse (Table 3): the maximum conductance in the absence of resource depletion g , the fraction of resources spent per spike U_{se} , the synaptic-current decay constant τ_d , the facilitation constant τ_f and the resource-recovery constant τ_r ([Moradi et al. 2022](#)). The same table also reports the connection probability P ; although Hippocampome.org supplies these probabilities *in vivo*, the reduced scale of the network made it necessary to raise them so that activity could be sustained without losing sparseness.

Cell	k (nS/mV)	a (ms ⁻¹)	b (nS)	d (pA)	C_m (pF)	V_r (mV)	V_t (mV)	V_{min} (mV)	V_{peak} (mV)
 mGC	0.45	0.003	24.48	50	38	-77.4	-44.9	-66.47	15.49
 iGC	0.139	0.002	-1.877	12.149	24.6	-63.66	-38.41	-48.2	83.5
 MC	1.5	0.004	-20.84	117	258	-63.67	-37.11	-47.98	28.29
 HIP	0.01	0.004	-2	40.52	58.7	-70	-50	-75	90
 BC	0.81	0.097	1.89	553	208	-61.02	-37.84	-36.23	14.08
 pCA3	0.79	0.008	-10	588	186	-63.2	-33.6	-38.87	35.86
 iCA3	1	0.004	-18.26	12	45	-57.51	-23.38	-47.56	18.45

Table 2 Izhikevich model parameters per cell type.

Three state variables define the synapse: the utilization of synaptic resources (U), their recovery (R) and the fraction held in the active state (A). All resources start available, so that $U_{t_0} = 0$, $R_{t_0} = 1$ and $A_{t_0} = 0$. They evolve according to:

$$\frac{dU}{dt} = \frac{-U}{\tau_f} + U_{se}(1 - U_-)\delta(\Delta t_i) \quad (4)$$

$$\frac{dR}{dt} = \frac{1 - R - A}{\tau_r} - U_+R_- \delta(\Delta t_i) \quad (5)$$

$$\frac{dA}{dt} = \frac{-A}{\tau_d} + U_+R_- \delta(\Delta t_i) \quad (6)$$

where δ is the Dirac delta, equal to 1 only when $\Delta t_i = t - t_i = 0$, that is, at the instant of a synaptic event t_i ; U_+ denotes the value of U just after the event and R_- the value of R just before it. The resulting synaptic current is:

$$I_{syn} = s \cdot A \cdot g \cdot (V_m - E) \quad (7)$$

where V_m is the postsynaptic membrane potential and E the reversal potential, set to $E_{inh} = -86$ mV for inhibitory synapses and $E_{exc} = 0$ mV for excitatory ones. The constant $s = 10$, applied uniformly to every synapse, is a scaling factor required by the reduced size of the network: with so few synapses relative to the rat hippocampus, the network would otherwise fall silent. The synaptic currents I_{syn} reaching a neuron are summed into the total current I of its membrane equation (Eq. 1).

2.4 Pattern separation

Pattern separation was quantified following [Kim and Lim \(2024b\)](#). To probe its dependence on input similarity and on adult neurogenesis, 11 variants of the network were compared: a control without neurogenesis ($N_{mGC} = 2000$) and 10 with neurogenesis ($N_{mGC} = 1900$, $N_{iGC} = 100$) at increasing levels of iGC excitability, as illustrated in Figure 2. This level, ranging from 0.1 to 1.0 in increments of 0.1, fixes the EC to iGC

Presynaptic	Postsynaptic	Connection	P (%)	g (nS)	τ_d (ms)	τ_r (ms)	τ_f (ms)	U
 EC	 mGC	Random	8	1.655	5.333	266.239	18.714	0.27
 EC	 iGC	Random	0.8–8*	1.655	5.333	266.239	18.714	0.27
 EC	 MC	Random	20	1.522	4.671	319.835	57.766	0.204
 EC	 BC	Random	25	1.406	3.849	144.415	48.2	0.214
 EC	 pCA3	Random	12	1.5	6.55	258.318	53.478	0.184
 EC	 iCA3	Random	12	2	3.602	257.468	35.904	0.21
 mGC	 MC	Random	40	2.713	5.347	98.583	27.479	0.151
 mGC	 HIPP	Random	10	1.805	5.181	362.814	48.986	0.15
 mGC	 BC	Lamellar	100	4.258	5.566	31.265	4.278	0.497
 mGC	 pCA3	Lamellar	80	1.5	6.657	278.286	78.584	0.155
 mGC	 iCA3	Lamellar	50	3	3.915	218.934	43.274	0.176
 iGC	 mGC	Random	10	0.524	3.915	218.934	43.274	0.176
 iGC	 MC	Random	40	2.713	5.347	98.583	27.479	0.151
 iGC	 HIPP	Random	10	1.805	5.181	362.814	48.986	0.15
 iGC	 BC	Lamellar	100	4.258	5.566	31.265	4.278	0.497
 iGC	 pCA3	Lamellar	80	1.5	6.657	278.286	78.584	0.155
 iGC	 iCA3	Lamellar	50	3	3.915	218.934	43.274	0.176
 MC	 mGC	Cross-lamellar	0.2	1.894	5.357	166.162	20.224	0.304
 MC	 iGC	Cross-lamellar	0.2	1.894	5.357	166.162	20.224	0.304
 MC	 MC	Cross-lamellar	5	0.068	4.257	149.329	71.642	0.245
 MC	 HIPP	Cross-lamellar	50	2.376	4.824	358.431	54.872	0.181
 MC	 BC	Cross-lamellar	100	1.996	3.396	117.365	69.316	0.255
 HIPP	 mGC	Random	30	3.202	8.935	159.143	10.396	0.278
 HIPP	 iGC	Random	10	3.202	8.935	159.143	10.396	0.278
 BC	 mGC	Lamellar	100	4.351	3.543	103.876	12.347	0.332
 BC	 iGC	Lamellar	10	4.351	3.543	103.876	12.347	0.332
 pCA3	 pCA3	Random	2.5	0.9	9.516	278.258	27.513	0.172
 pCA3	 MC	Lamellar	10	2.035	4.297	359.116	40.457	0.236
 pCA3	 iCA3	Random	20	2	4.525	275.604	23.321	0.189
 iCA3	 pCA3	Random	30	2	7.793	416.282	20.63	0.203

Table 3 Parameters of the synapses between neuronal populations. Random connections occur between all cells of both populations; lamellar connections occur between cells of the same lamella; cross-lamellar connections occur between cells of one lamella and all others. The connection probability P refers to the percentage of connections between neuronal populations according to the connection condition. * iGCs were simulated with multiple different EC→iGC excitability levels.

connection probability as a fraction relative to that of the mGCs: with 1.0 matching the mGC connection probability, and 0.1 meaning 10% of that probability, and so represents the still-incomplete dendritic integration of the immature cells. Each

variant was presented with 30 sets of 10 input patterns, a set being one randomly generated reference pattern together with 9 derivatives of it graded from 90% down to 10% similarity. An input pattern is a binary vector of length $N_{EC} = 400$ encoding the state of each EC neuron, its active neurons firing as Poisson processes at 40 Hz and exactly 10% (40 neurons) being active in every pattern. A pattern of similarity $X\%$ was obtained by keeping a random fraction of $X\%$ of the reference pattern's active neurons and moving the remaining $(100 - X)\%$ to previously inactive neurons, which preserves the 10% activity level.

Each pattern is simulated for 1300 ms of simulation time, with the first 300 ms discarded while the network stabilizes. The output pattern of a neuronal population, whether mGCs, iGCs, or pCA3 cells, consists of a binary vector with value 1 for the index of each neuron that spiked at least once during the 1000 ms interval.

Separation was then read from the overlap between the activity patterns at the input (EC cells) and the output (GCs or pCA3 cells). A pattern is a binary vector of length N , the number of cells in the population, whose entries are 1 for neurons that fired at least once during the 1000 ms window and 0 otherwise. For a pair of patterns $A^{(l)}$ and $B^{(l)}$, with $l \in \{in, out\}$ for input and output, the pattern distance $D_p^{(l)}$ is defined as:

$$D_p^{(l)} = \frac{O^{(l)}}{D_a^{(l)}} \quad (8)$$

Here $O^{(l)}$ is the degree of orthogonalization and $D_a^{(l)}$ the mean activation of the pair, itself the arithmetic mean of the activation degrees of $A^{(l)}$ and $B^{(l)}$:

$$D_a^{(l)} = \frac{D_a^{(A^{(l)})} + D_a^{(B^{(l)})}}{2} \quad (9)$$

the activation degree of a pattern being its fraction of active neurons. The orthogonalization $O^{(l)}$, which gauges the dissimilarity between the patterns, follows from the Pearson correlation coefficient $\rho^{(l)}$:

$$O^{(l)} = \frac{1 - \rho^{(l)}}{2} \quad (10)$$

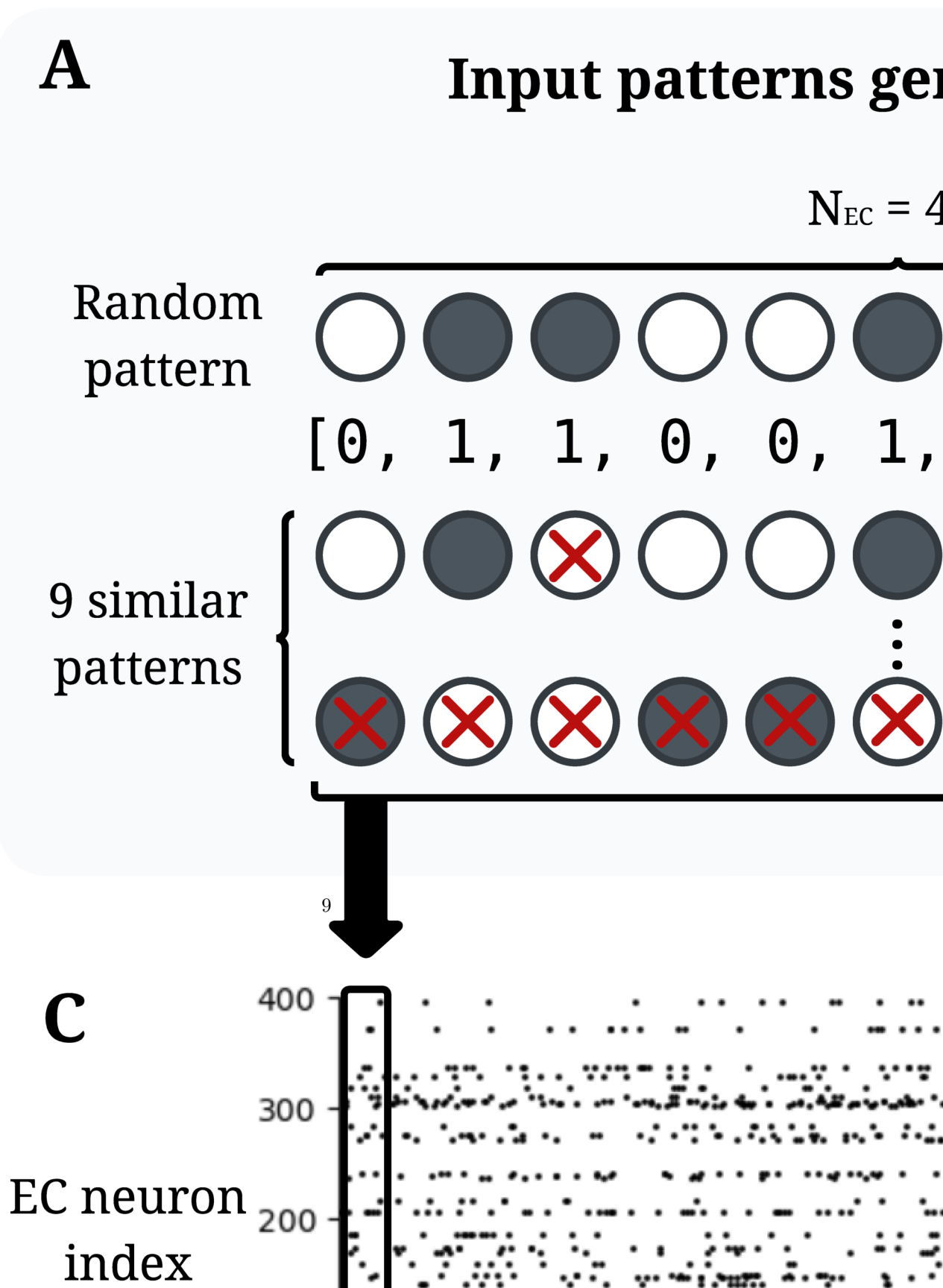
Writing $\{a_i^{(l)}\}$ and $\{b_i^{(l)}\}$ ($i = 1, \dots, N_l$) for the binary states of the i -th cell in $A^{(l)}$ and $B^{(l)}$ ($l \in \{in, out\}$), this coefficient is:

$$\rho^{(l)} = \frac{\sum_{i=1}^{N_l} \Delta a_i^{(l)} \cdot \Delta b_i^{(l)}}{\sqrt{\sum_{i=1}^{N_l} (\Delta a_i^{(l)})^2} \sqrt{\sum_{i=1}^{N_l} (\Delta b_i^{(l)})^2}} \quad (11)$$

with $\Delta a_i^{(l)} = a_i^{(l)} - \langle a^{(l)} \rangle$, $\Delta b_i^{(l)} = b_i^{(l)} - \langle b^{(l)} \rangle$ and $\langle \dots \rangle$ the mean over the whole population. Since $\rho^{(l)}$ ranges over $[-1, 1]$, the orthogonalization $O^{(l)}$ ranges over $[0, 1]$ (Eq. 10).

From the input ($D_p^{(in)}$) and output ($D_p^{(out)}$) distances, the degree of pattern separation $\mathcal{S}_D$ is taken as their ratio:

Fig. 2 Diagram of the input and output pattern generation and recall protocol. **A.** Input patterns are generated in sets of 10 patterns of size $N_{EC} = 400$; the first is generated randomly and the remaining ones are generated from it at different levels of similarity by randomly activating and deactivating neurons. **B.** 11 model variants are simulated: one without neurogenesis and 10 with neurogenesis at different connectivity levels, from 100% down to 10% connectivity. **C.** Spike trace of EC neurons during a simulation; each generated pattern corresponds to the activation of EC neurons, with active neurons spiking at a frequency of 40 Hz; each of the 11 models is exposed to 30 sets of 10 input patterns, totaling 300 simulations of 1300 ms each. **D.** Spike trace of mGCs during a simulation; the first 300 ms are discarded while the network stabilizes; the remainder corresponds to the output pattern. **E.** The output pattern, illustrated here for the mGC population, consists of a binary vector with value 1 for the index of each neuron that spiked at least once during the 1000 ms interval.



$$\mathcal{S}_D = \frac{D_p^{(out)}}{D_p^{(in)}} \quad (12)$$

A value $\mathcal{S}_D > 1$ means that the output patterns are more distinct than the inputs, i.e. pattern separation, whereas $\mathcal{S}_D < 1$ means the opposite, a convergence in which the outputs grow more alike.

2.5 iGC activation

3 Results

4 Discussion

Supplementary information. Online Resource 1 contains a table comparing model firing rates and population sparsity, per cell type, against a compilation of *in vivo* electrophysiological data from the literature.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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